

Assignment-2 (Fourier Series)

Q1. State Dirichlet's conditions for expansion of $f(x)$ in Fourier series.

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx$$

- (i) $f(x)$ is periodic, single value and finite.
- (ii) $f(x)$ has no. of finite discontinuity in any one period.
- (iii) $f(x)$ has a (period) finite no. of maxima and minima.

Q2. Obtain the Fourier series to represent $f(x) = \frac{1}{4}(\pi - x)^2$ in the interval $0 \leq x \leq 2\pi$.

Hence, obtain the following relations:

$$(i) \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \dots = \frac{\pi^2}{6}$$

$$(ii) \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots = \frac{\pi^2}{12}$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx$$

$$a_0 = \frac{1}{\pi} \int_0^{2\pi} f(x) dx = \frac{1}{4\pi} \int_0^{2\pi} (\pi - x)^2 dx$$

$$= \frac{1}{4\pi} \left[\frac{(\pi - x)^3}{-3} \right]_0^{2\pi}$$

$$= \frac{1}{4\pi} \left(\frac{(\pi - 2\pi)^3}{-3} - \frac{(\pi - 0)^3}{-3} \right)$$

$$= \frac{1}{4\pi} \left(\frac{-\pi^3}{-3} + \frac{\pi^3}{3} \right) = \left[\frac{\pi^2}{6} \right]$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx \, dx = \frac{1}{4\pi} \int_0^{2\pi} (\pi-x)^2 \cos nx \, dx$$

$$= \frac{1}{4\pi} \left[(\pi-x)^2 \frac{\sin nx}{n} + \int 2(\pi-x) \frac{\sin nx}{n} \right]_0^{2\pi}$$

$$= \frac{1}{4n\pi} \left[(\pi-x)^2 \sin nx + 2 \left[(\pi-x) \frac{(-\cos nx)}{n} - \int (-1) \frac{(-\cos nx)}{n} \right] \right]_0^{2\pi}$$

$$= \frac{1}{4n\pi} \left[(\pi-x)^2 \sin nx + \frac{2}{n} \left[-(\pi-x) \cos nx - \frac{\sin nx}{n} \right] \right]_0^{2\pi}$$

$$= \frac{1}{4n\pi} \left[0 + \frac{2}{n} \left[\pi - \left(\frac{2}{n} (-\pi) \right) \right] \right]$$

$$= \frac{1}{4n\pi} \left(\frac{4\pi}{n} \right) = \boxed{\frac{1}{n^2}}$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx \, dx$$

$$b_n = \frac{1}{4\pi} \int_0^{2\pi} (\pi-x)^2 \sin nx \, dx$$

$$= \frac{1}{4\pi} \left[(\pi-x)^2 \frac{(-\cos nx)}{n} - \int 2(\pi-x) \frac{(-\cos nx)}{n} \right]_0^{2\pi}$$

$$= \frac{1}{4n\pi} \left[-(\pi-x)^2 \frac{\cos nx}{n} - 2 \left[(\pi-x) \frac{\sin nx}{n} + \int \frac{\sin nx}{n} \right] \right]_0^{2\pi}$$

$$= \frac{1}{4n\pi} \left[-(\pi-x)^2 \cos nx - \frac{2}{n} \left[(\pi-x) \sin nx - \frac{\cos nx}{n} \right] \right]_0^{2\pi}$$

$$= \frac{1}{4n\pi} \left[-(\pi-2\pi)^2 \cos 2n\pi - \frac{2}{n} \left[(\pi-2\pi) \sin 2n\pi - \frac{\cos 2n\pi}{n} \right] \right] - \left[-(\pi-0)^2 \cos 0 - \frac{2}{n} \left[(\pi-0) \sin 0 - \frac{\cos 0}{n} \right] \right]$$

$$= \frac{1}{4n\pi} \left[-(\pi-2\pi)^2 \cos 2n\pi - \frac{2}{n} \left[(\pi-2\pi) \sin 2n\pi - \frac{\cos 2n\pi}{n} \right] \right] - \left[-(\pi-0)^2 \cos 0 - \frac{2}{n} \left[(\pi-0) \sin 0 - \frac{\cos 0}{n} \right] \right]$$

$$= \frac{1}{4n\pi} \left[-\pi^2 - \frac{2}{n} \left[0 - \frac{1}{n} \right] - \left(-\pi^2 - \frac{2}{n} \left(0 - \frac{1}{n} \right) \right) \right]$$

$$= \frac{1}{4n\pi} \left[-\pi^2 + \frac{2}{n^2} + \pi^2 - \frac{2}{n^2} \right] = \boxed{0}$$

$$\boxed{b_n = 0}$$

$$\boxed{f(x) = \frac{\pi^2}{12} + \sum_{n=1}^{\infty} \frac{1}{n^2} \cos nx}$$

$$\textcircled{1} \quad \frac{1}{4} (\pi - x)^2 = \frac{\pi^2}{12} + \left(\frac{\cos x}{1^2} + \frac{\cos 2x}{2^2} + \frac{\cos 3x}{3^2} + \dots \right)$$

$$\frac{\cos x}{1^2} + \frac{\cos 2x}{2^2} + \frac{\cos 3x}{3^2} + \dots = \frac{1}{4} (\pi - x)^2 - \frac{\pi^2}{12}$$

$$(i) \quad \text{Put } x=0$$

$$\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots = \frac{1}{4} \pi^2 - \frac{\pi^2}{12}$$

$$\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots = \frac{\pi^2}{6}$$

$$(ii) \quad \text{Put } x=\pi$$

$$\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots = \cancel{\frac{1}{4}} \pi^2 - \frac{\pi^2}{12}$$

$$\textcircled{3} \quad \text{Expand } f(x) = |x| \text{ in a fourier series in } (-\pi, \pi)$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx$$

$$|x| = \begin{cases} x & \pi > x > 0 \\ -x & -\pi < x < 0 \end{cases}$$

Check even/odd function

If $f(x) = f(x)$ then the fn is even

" $f(-x) = -f(x)$ then the fn is odd

$$f(-x) = |-x| = x = f(x)$$

the function is even so, $b_n = 0$

As fn is even, so,

$$a_0 = \frac{2}{\pi} \int_0^{\pi} x dx$$

$$= \frac{2}{\pi} \left(\frac{x^2}{2} \right)_0^{\pi} = \frac{2}{\pi} \left(\frac{\pi^2}{2} \right)$$

$$= \boxed{\pi}$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} x^n \cos nx dx$$

$$= \frac{2}{\pi} \left[x \frac{\sin nx}{n} - \int (1) \frac{\sin nx}{n} dx \right]_0^{\pi}$$

$$= \frac{2}{n\pi} \left[x \sin nx + \frac{\cos nx}{n} \right]_0^{\pi}$$

$$= \frac{2}{n\pi} \left(0 + \frac{(-1)^n}{n} - \frac{(+1)^0}{n} \right)$$

$$= \frac{2}{n^2\pi} ((-1)^n - 1)$$

$$|x| = \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{2}{n^2\pi} ((-1)^n - 1) \cos nx$$

eg

$$|x| = \frac{\pi}{2} + \frac{2}{\pi} (-2)^{\cos n} + \frac{2}{4\pi} (1-1)^{\cos 2n} + \frac{2}{9\pi} (-1-1)^{\cos 3n} + \dots$$

$$|x| = \frac{\pi}{2} + \frac{4 \cos n}{\pi} - \frac{4 \cos 3n}{9\pi} + \dots$$

$$|x| = \frac{\pi}{2} - \frac{4}{\pi} \left(\cos x + \frac{\cos 3x}{3^2} + \dots \right)$$

Q:- 4 Expand $f(x) = x \sin x$, $0 < x < 2\pi$ as a fourier

in $(-\pi, \pi)$

Put $f(x) = f(-x)$
 $f(-x) = (-x) \sin(-x)$
 $f(-x) = f(x)$
 So, $b_n = 0$

$$\begin{aligned} a_0 &= \frac{1}{\pi} \int_0^{2\pi} x \sin x \, dx = \frac{1}{\pi} \left[-x \cos x + \int \cos x \, dx \right]_0^{2\pi} \\ &= \frac{1}{\pi} \left[-x \cos x + \sin x \right]_0^{2\pi} \\ &= \frac{1}{\pi} [-2\pi + 0] = -2 \end{aligned}$$

$$\begin{aligned} a_n &= \frac{1}{\pi} \int_0^{2\pi} x \sin x \cos nx \, dx \\ &= \frac{1}{2\pi} \int_0^{2\pi} x [\sin(x+nx) - \sin(x-nx)] \, dx \\ &= \frac{1}{2\pi} \left[\int_0^{2\pi} x \sin(x+nx) \, dx - \int_0^{2\pi} x \sin(x-nx) \, dx \right] \\ &= \frac{1}{2\pi} \left[-x \frac{\cos(nx+x)}{1+n} + \frac{\cos(nx-x)}{1-n} \right] + \left[\frac{\cos(nx+x)}{1+n} + \frac{\cos(x-nx)}{1-n} \right]_0^{2\pi} \\ &= \frac{1}{2\pi} \left[-x \left(\frac{\cos(nx+x)}{1+n} + \frac{\cos(x-nx)}{1-n} \right) + \frac{\sin(nx+x)}{(1+n)^2} + \frac{\sin(x-nx)}{(1-n)^2} \right]_0^{2\pi} \\ &= \frac{1}{2\pi} \left[-2\pi \left[\frac{1}{1+n} + \frac{1}{1-n} \right] \right] \end{aligned}$$

$$= - \left(\frac{1-x + 1+x}{(1+n)(1-n)} \right)$$

$$a_n = \left[\frac{1}{n^2-1} \right]$$

$$b_n = \frac{1}{\pi} \int_0^{2\pi} x \sin nx \sin nx \, dx = \frac{1}{2\pi} \int_0^{2\pi} x \left(\cos(x-nx) - \cos(x+nx) \right) dx$$

$$= \frac{1}{2\pi} \left[x \left(\frac{\sin(x-nx)}{1-n} - \frac{\sin(x+nx)}{1+n} \right) - \int \left(\frac{\sin(x-nx)}{1-n} - \frac{\sin(x+nx)}{1+n} \right) dx \right]_0^{2\pi}$$

$$= \frac{1}{2\pi} \left[x \left(\frac{\sin(x-nx)}{1-n} - \frac{\sin(x+nx)}{1+n} \right) + \frac{\cos(n-n)}{(1-n)^2} \left(\frac{\sin(x+nx)}{1+n} \right) - \frac{\cos(x-nx)}{(1+n)^2} \right]_0^{2\pi}$$

$$= \frac{1}{2\pi} \left[\frac{1}{(1-n)^2} - \frac{1}{(1+n)^2} - \frac{1}{(1-n)^2} + \frac{1}{(1+n)^2} \right]$$

$$=$$

6) Find fourier series of $f(x) = x^3$ in $(-\pi, \pi)$

Soln:-

$$f(-x) = (-x)^3$$

$$f(-x) = -x^3$$

$f(-x) = -f(x)$, As the f^n is odd

$$\text{so, } a_n = 0, a_0 = 0$$

$$b_n = \frac{2}{\pi} \int_0^{\pi} f(x) \sin nx \, dx$$

$$= \frac{2}{\pi} \int_0^{\pi} x^3 \sin nx \, dx$$

$$= \frac{2}{\pi} \left[\frac{-x^3 \cos nx}{n} + \int \frac{3x^2 \cos nx}{n} \, dx \right]_0^{\pi}$$

$$= \frac{2}{n\pi} \left[\frac{-x^3 \cos nx}{n} + 3 \left[\frac{x^2 \sin nx}{n} - \int \frac{2x \sin nx}{n} \, dx \right] \right]$$

$$= \frac{2}{n\pi} \left[\frac{-x^3 \cos nx}{n} + \frac{3}{n} \left[x^2 \sin nx + 2 \left[\frac{x \cos nx}{n} - \int \frac{\cos nx}{n} \, dx \right] \right] \right]$$

$$= \frac{2}{n\pi} \left[\frac{-x^3 \cos nx}{n} + \frac{3}{n} \left[x^2 \sin nx + 2 \left[\frac{x \cos nx}{n} - \frac{\sin nx}{n^2} \right] \right] \right]$$

$$= \frac{2}{n\pi} \left[\frac{-\pi^3 (-1)^n}{n} + \frac{3}{n} \left[0 + 2 \left(\frac{\pi (-1)^n}{n} - 0 \right) \right] \right]$$

$$= \frac{2}{n\pi} \left[\frac{-\pi^3 (-1)^n}{n} + \frac{6\pi (-1)^n}{n^2} \right]$$

$$b_n = \frac{2}{n} \left[\frac{6(-1)^n}{n^2} - \frac{\pi^2 (-1)^n}{n} \right]$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx$$

$$f(x) = \sum_{n=1}^{\infty} \left(\frac{6(-1)^n}{n^3} - \frac{\pi^2 (-1)^n}{n} \right) \sin nx$$

⑥ Find the Fourier series for the following function

$$f(x) = \begin{cases} -1 & ; -\pi < x < -\pi/2 \\ 0 & ; -\pi/2 < x < \pi/2 \\ 1 & ; \pi/2 < x < \pi \end{cases}$$

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{\pi} \left[\int_{-\pi}^{-\pi/2} -1 dx + \int_{-\pi/2}^{\pi/2} 0 dx + \int_{\pi/2}^{\pi} 1 dx \right]$$

$$= \left[-x \right]_{-\pi}^{-\pi/2} + 0 + \left[x \right]_{\pi/2}^{\pi}$$

$$= \left[\frac{\pi}{2} - \pi \right] + \left[\pi - \frac{\pi}{2} \right]$$

$$= -\frac{\pi}{2} + \frac{\pi}{2} = 0$$

$$a_n = \frac{1}{\pi} \left[\int_{-\pi}^{\pi} f(x) \cos nx dx \right]$$

$$= \frac{1}{\pi} \left[\int_{-\pi}^{-\pi/2} -\cos nx dx + \int_{-\pi/2}^{\pi/2} 0 \cos nx dx + \int_{\pi/2}^{\pi} \cos nx dx \right]$$

$$= \frac{1}{\pi} \left[\left[-\frac{\sin nx}{n} \right]_{-\pi}^{-\pi/2} + 0 + \left[\frac{\sin nx}{n} \right]_{\pi/2}^{\pi} \right]$$

$$= \frac{1}{\pi} \left[\left[\frac{+1}{n} - 0 \right] + 0 + \left[\frac{\sin n\pi}{n} - \frac{1}{n} \right] \right]$$

$$= \frac{1}{\pi} \left[\frac{1}{n} - \frac{1}{n} \right] = 0$$

$$b_n = \frac{1}{\pi} \left[\int_{-\pi}^{\pi} f(x) \sin nx dx \right]$$

$$= \frac{1}{\pi} \left[\int_{-\pi}^{-\pi/2} -\sin nx dx + \int_{-\pi/2}^{\pi/2} 0 dx + \int_{\pi/2}^{\pi} \sin nx dx \right]$$

$$= \frac{1}{\pi} \left[\left[\frac{\cos nx}{n} \right]_{-\pi}^{-\pi/2} + 0 + \left[-\frac{\cos nx}{n} \right]_{\pi/2}^{\pi} \right]$$

$$= \frac{1}{\pi} \left[\frac{-(-1)^n}{n} - \frac{(-1)^n}{n} \right]$$

$$b_n = \left[\frac{-2(-1)^n}{n\pi} \right]$$

$$f(x) = \sum_{n=1}^{\infty} \frac{(-2)(-1)^n}{n\pi} \sin nx$$

$$f(x) = \frac{(-2)(-1)}{\pi} \sin x + \frac{(-2)(-1)^2}{2\pi} \sin 2x + \frac{(-2)(-1)^3}{3\pi} \sin 3x + \dots$$

$$f(x) = \frac{2}{\pi} \left[\sin x + \frac{\sin 2x}{2} + \frac{\sin 3x}{3} + \dots \right]$$

7. Obtain the Fourier series for the function $f(x)$ given by.

$$f(x) = \begin{cases} 1 + \frac{2x}{\pi}, & -\pi \leq x \leq 0 \\ 1 - \frac{2x}{\pi}, & 0 \leq x \leq \pi \end{cases}$$

Hence deduce that $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots = \frac{\pi^2}{8}$

$$-\pi \leq x \leq 0 \quad \& \quad 1 + \frac{2x}{\pi}$$

$$\pi \geq x \geq 0 \quad , \quad 1 - \frac{2x}{\pi}$$

$$0 \leq x \leq \pi \quad , \quad 1 - \frac{2x}{\pi}$$

$$0 \geq x \geq -\pi \quad , \quad 1 + \frac{2x}{\pi}$$

So, the function is even,
 $b_n = 0$

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$$\begin{aligned} \frac{a_0}{2} &= \frac{2}{\pi} \int_0^{\pi} f(x) dx = \frac{2}{\pi} \int_0^{\pi} \left(1 - \frac{2x}{\pi}\right) dx \\ &= \frac{2}{\pi} \left[x - \frac{2x^2}{2\pi} \right]_0^{\pi} = \frac{2}{\pi} \left[\pi - \frac{\pi^2}{\pi} - 0 - 0 \right] \\ &= \boxed{0} \end{aligned}$$

$$\begin{aligned} a_n &= \frac{2}{\pi} \left[\int_0^{\pi} \left(1 - \frac{2x}{\pi}\right) \cos nx \, dx \right] \\ &= \frac{2}{\pi} \left[\left(1 - \frac{2x}{\pi}\right) \frac{\sin nx}{n} + \int \frac{2}{\pi} \frac{\sin nx}{n} dx \right]_0^{\pi} \\ &= \frac{2}{n\pi} \left[\left(1 - \frac{2x}{\pi}\right) \sin nx + \frac{2}{\pi} \left[\frac{\cos nx}{n} \right] \right]_0^{\pi} \\ &= \frac{2}{n\pi} \left[-\sin n\pi - \frac{2}{n\pi} (-1)^n - 0 + \frac{2}{n\pi} \right] \\ &= \frac{2}{n\pi} \left(\frac{2}{n\pi} (1 - (-1)^n) \right) \end{aligned}$$

$$a_n = \frac{4}{n^2 \pi^2} (1 - (-1)^n)$$

$$f(x) = 4 \sum_{n=1}^{\infty} \frac{1}{n^2 \pi^2} (1 - (-1)^n) \cos nx$$

$$f(x) = \frac{4}{\pi^2} \left[\frac{2 \cos x}{1^2} + \frac{2 \cos 3x}{3^2} + \frac{2 \cos 5x}{5^2} + \dots \right]$$

$$\left(1 - \frac{2x}{\pi}\right) \frac{x \pi^2}{4 \times 2} = \frac{\cos x}{1^2} + \frac{\cos 3x}{3^2} + \frac{\cos 5x}{5^2} + \dots$$

Put $x=0$

$$\frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots$$

Change of interval

Q: 8 Find Fourier expansion for the function

$$f(x) = x - x^2, \quad -1 < x < 1$$

$$-l < x < l$$

$$\text{So, } l=1$$

$$a_0 = \frac{1}{l} \int_{-l}^l f(x) dx$$

$$= \frac{1}{2} \int_{-1}^1 (x - x^2) dx$$

$$= \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_{-1}^1 = \left[\frac{-2}{3} \right]$$

$$a_n = \frac{1}{l} \int_{-l}^l f(x) \cos \frac{n\pi x}{l} dx$$

$$= \frac{1}{2} \int_{-1}^1 (x - x^2) \cos n\pi x dx$$

$$= \int_{-1}^1 x \cos n\pi x dx - \int_{-1}^1 x^2 \cos n\pi x dx$$

$$= 0 - 2 \int_0^1 x^2 \cos n\pi x dx$$

$$= -2 \left[\frac{x^2 \sin n\pi x}{n\pi} - \int \frac{2x \sin n\pi x}{n\pi} \right]_0^1$$

$$= \frac{-2}{n\pi} \int x^2 \sin n\pi x + 2 \left[\frac{x \cos n\pi x}{n\pi} - \int \frac{\cos n\pi x}{n\pi} \right]$$

$$= \frac{-2}{n\pi} \left[x^2 \sin n\pi x + \frac{2}{n\pi} \left(x \cos n\pi x - \frac{\sin n\pi x}{n\pi} \right) \right]_0^1$$

$$= \frac{-2}{n\pi} \left[0 + \frac{2}{n\pi} ((-1)^n - 0) - 0 \right]$$

$$a_n = \frac{-4}{(n\pi)^2} (-1)^n$$

$$\begin{aligned}
 b_n &= \frac{1}{l} \int_{-l}^{l+2\pi} f(x) \frac{\sin nx}{l} dx \\
 &= \int_{-1}^1 (x-x^2) \sin n\pi x dx \\
 &= \int_{-1}^1 x \sin n\pi x dx - \int_{-1}^1 x^2 \sin n\pi x dx \\
 &= \left[-x \frac{\cos n\pi x}{n\pi} + \int \frac{\cos n\pi x}{n\pi} \right]_0^1 - 0 \\
 &= 2 \left[-x \frac{\cos n\pi x}{n\pi} + \frac{\sin n\pi x}{(n\pi)^2} \right]_0^1 \\
 &= 2 \left[-\frac{(-1)^n}{n\pi} + 0 - 0 - 0 \right]
 \end{aligned}$$

$$b_n = \frac{-2(-1)^n}{n\pi}$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos n\pi x + \sum_{n=1}^{\infty} b_n \sin n\pi x$$

$$f(x) = \frac{-1}{3} - \sum_{n=1}^{\infty} \frac{4(-1)^n}{(n\pi)^2} \cos n\pi x - \sum_{n=1}^{\infty} \frac{2(-1)^n}{n\pi} \sin n\pi x$$

Q:-9 Obtain fourier series for function $f(x) = \begin{cases} \pi x & 0 \leq x \leq 1 \\ \pi(2-x) & 1 \leq x \leq 2 \end{cases}$

$$a_0 = \frac{1}{l} \int_{-l}^{l+2\pi} f(x) dx$$

$$a_0 = \int_0^2 f(x) dx = \int_0^1 \pi x dx + \int_1^2 \pi(2-x) dx \quad [l=1]$$

$$= \left[\frac{\pi x^2}{2} \right]_0^1 + \pi \left[2x - \frac{x^2}{2} \right]_1^2$$

$$= \frac{\pi}{2} - 0 + \pi \left(4 - \frac{4}{2} - 2 + \frac{1}{2} \right)$$

$$= \frac{\pi}{2} + \frac{\pi}{2} = \pi$$

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$$a_n = \frac{1}{2} \int_0^2 f(x) \cos n\pi x dx = \int_0^1 \pi x \cos n\pi x dx + \int_1^2 \pi(2-x) \cos n\pi x dx$$

$$= \left[\frac{\pi x \sin n\pi x}{n\pi} - \int \frac{\sin n\pi x}{n\pi} dx \right]_0^1 + \left[\frac{\pi(2-x) \sin n\pi x}{n\pi} + \int \pi \sin n\pi x dx \right]_1^2$$

$$= \left[\frac{x \sin n\pi x}{n} + \frac{\cos n\pi x}{n^2 \pi} \right]_0^1 + \left[(2-x) \sin n\pi x - \frac{\cos n\pi x}{n^2 \pi} \right]_1^2$$

$$= \left(0 + \frac{(-1)^n}{n^2 \pi} - \frac{1}{n^2 \pi} \right) + \left(\frac{-1}{n^2 \pi} + \frac{(-1)^n}{n^2 \pi} \right)$$

$$a_n = 2 \left[\frac{(-1)^n}{n^2 \pi} - \frac{1}{n^2 \pi} \right]$$

$$b_n = \frac{1}{2} \int_0^2 f(x) \sin n\pi x dx = \int_0^1 \pi x \sin n\pi x dx + \int_1^2 \pi(2-x) \sin n\pi x dx$$

$$= \left[-\frac{\pi x \cos n\pi x}{n\pi} + \int \frac{\cos n\pi x}{n\pi} dx \right]_0^1 + \left[-(2-x) \cos n\pi x - \int \cos n\pi x dx \right]_1^2$$

$$= \left[-\frac{x \cos n\pi x}{n} + \frac{\sin n\pi x}{n^2 \pi} \right]_0^1 + \left[-(2-x) \cos n\pi x - \frac{\sin n\pi x}{n^2 \pi} \right]_1^2$$

$$= \frac{-(-1)^n}{n} \neq 0 - 0 + \frac{(-1)^n}{n} - 0$$

$$= 0$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos n\pi x + \sum_{n=1}^{\infty} b_n \sin n\pi x$$

$$= \frac{\pi}{2} + \sum_{n=1}^{\infty} 2 \left(\frac{(-1)^n}{n^2 \pi} - \frac{1}{n^2 \pi} \right) \cos n\pi x + 0$$

$$= \frac{\pi}{2} + 2 \left(\frac{(-2) \cos \pi x}{2^2 \pi} - \frac{2 \cos 3\pi x}{2^2 \pi} - \frac{2 \cos 5\pi x}{2^2 \pi} \right)$$

$$f(x) = \frac{\pi}{2} - \frac{4}{\pi^2} \left[\frac{\cos \pi x}{1^2} + \frac{\cos 3\pi x}{2^2} + \frac{\cos 5\pi x}{3^2} + \dots \right]$$

(10) Develop $f(x)$ in a Fourier series in the interval $(0, 2)$
 in $f(x) = \begin{cases} x & 0 < x < 1 \\ 0 & 1 < x < 2 \end{cases}$

Overall limit is $0 < x < 2$, $0 < x < 2$

$$a_0 = \frac{1}{l} \int_0^{2l} f(x) dx = \int_0^1 x dx + \int_1^2 0 dx$$

$$= \left[\frac{x^2}{2} \right]_0^1 = \left[\frac{1}{2} \right]$$

$$a_n = \frac{1}{l} \int_0^{2l} f(x) \cos n\pi x dx = \int_0^1 x \cos n\pi x dx + \int_1^2 0 \cos n\pi x dx$$

$$= \left[\frac{x \sin n\pi x}{n\pi} - \int \frac{\sin n\pi x}{n\pi} dx \right]_0^1 + 0$$

$$= \left[\frac{x \sin n\pi x}{n\pi} + \frac{\cos n\pi x}{(n\pi)^2} \right]_0^1$$

$$= \left[0 + \frac{(-1)^n}{(n\pi)^2} - 0 - \frac{1}{(n\pi)^2} \right]$$

$$= \left[\frac{(-1)^n}{(n\pi)^2} - \frac{1}{(n\pi)^2} \right] \begin{cases} a_n = 0, n = \text{even} \\ a_n \neq 0, n = \text{odd} \end{cases}$$

$$b_n = \frac{1}{l} \int_0^{2l} f(x) \sin n\pi x dx = \int_0^1 x \sin n\pi x dx + \int_1^2 0 \sin n\pi x dx$$

$$= \left[\frac{-x \cos n\pi x}{n\pi} + \int \frac{\cos n\pi x}{n\pi} dx \right]_0^1 + 0$$

$$= \left[\frac{-x \cos n\pi x}{n\pi} + \frac{\sin n\pi x}{(n\pi)^2} \right]_0^1 = \left[\frac{-(-1)^n}{n\pi} + 0 + \frac{1}{n\pi} - 0 \right]$$

$$b_n = \left[\frac{1}{n\pi} - \frac{(-1)^n}{n\pi} \right]$$

$$\begin{cases} n = \text{odd}, a_n \neq 0 \\ n = \text{even}, a_n = 0 \end{cases}$$

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$$f(x) = \frac{1}{4} + \sum_{n=1}^{\infty} \left(\frac{(-1)^n}{(n\pi)^2} - \frac{1}{(n\pi)^2} \right) \cos n\pi x + \sum_{n=1}^{\infty} \left(\frac{1}{n\pi} - \frac{(-1)^n}{n\pi} \right) \sin n\pi x$$

$$f(x) = \frac{1}{4} + \frac{2}{\pi^2} \left[\cos \pi x + \frac{\cos 3\pi x}{3^2} + \dots \right] + \frac{1}{\pi} \left[\sin \pi x - \frac{\sin 2\pi x}{2} + \frac{\sin 3\pi x}{3} - \dots \right]$$

Half Range Series:-

(ii) Expand $\pi x - x^2$ in a half-range sine series in the interval $(0, \pi)$ upto the first three terms.

Half range sine series

$$b_n = \frac{2}{\pi} \int_0^{\pi} f(x) \sin nx \, dx = \frac{2}{\pi} \int_0^{\pi} (\pi x - x^2) \sin nx \, dx$$

$$b_n = \frac{2}{\pi} \left[\int_0^{\pi} (\pi x - x^2) \frac{(-\cos nx)}{n} \, dx + \int_0^{\pi} (\pi - 2x) \frac{\cos nx}{n} \, dx \right]$$

$$= \frac{2}{n\pi} \left[-(\pi x - x^2) \frac{\cos nx}{n} + \int_0^{\pi} (\pi - 2x) \frac{\sin nx}{n} \, dx + \int_0^{\pi} 2 \frac{\sin nx}{n} \, dx \right]$$

$$= \frac{2}{n\pi} \left[-(\pi x - x^2) \cos nx + (\pi - 2x) \frac{\sin nx}{n} + \frac{2 \cos nx}{n^2} \right]_0^{\pi}$$

$$= \frac{2}{n\pi} \left[\frac{-2(-1)^n}{n^2} + \frac{2}{n^2} \right] = \frac{4}{n^3\pi} [1 - (-1)^n]$$

$$= \frac{8}{n^3\pi}, \quad n = \text{odd}$$

$$= 0, \quad n = \text{even}$$

Sine series,

$$f(x) = \sum_{n=1}^{\infty} b_n \sin nx = \sum_{n=1}^{\infty} \left(\frac{4}{n^3\pi} (1 - (-1)^n) \right) \sin nx$$

$$f(x) = \frac{8}{\pi} \left[\frac{\sin x}{1^3} + \frac{\sin 3x}{3^3} + \frac{\sin 5x}{5^3} + \dots \right]$$

(12)

Obtain a half-range cosine series for

$$f(x) = \begin{cases} Kx & \text{for } 0 < x < l/2 \\ K(l-x) & \text{for } l/2 < x < l \end{cases}$$

Deduce the sum of the series $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots = \frac{\pi^2}{8}$